CECH GLUEING OF DERIVED-CATEGORY OBJECTS ALONG A COVERING UNDER NEGATIVE-EXT AND COHOMOLOGICAL-BOUND VANISHING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the '(details omitted)' at source line 5796 covers the cocycle condition for the family rho_a: pulling back the cocycle identities of the rho_{ij} (condition (1)) along the induced maps to the triple fibre products U_i x_X U_j x_X U_k - an auxiliary pullback bookkeeping with all data defined in-span; the author's alternative 'second proof' (source lines 5830-5859), a sketch via the Cech hypercovering that ends with an explicitly omitted compatibility verification, lies outside the counted span and is not part of this card; the first sentence of the counted proof ('We urge the reader to look at the second proof in stead') is the author's framing, not an omission.

Editorial note. The label lemma-bbd-glueing-cech names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The general glueing theorem for locally given objects. Same chapter, \label{lemma-bbd-glueing}. Lines 5807-5828: applied with D = C and f the identity morphism; its assumptions (1), (2)(a)-(c), (3), (4) are verified.
- The locally-given situation for a covering. Same chapter, Situation \label{situation-locally-given}. Lines 5802-5805: the data (B, E_U, rho_a) are an instance of it.
- Derived pullback along morphisms of ringed sites. Standard functoriality of derived pullback a^*. Lines 5788-5800: E_U = a_U^* E_{i(U)} and the isomorphisms rho_a from the rho_{ij}.
- Vanishing of negative self-Exts descends along restriction. Functoriality of SheafExt under derived pullback. Lines 5819-5821: negative Ext vanishing for E_i implies the same for every restriction, feeding assumption (3).

Source reference. Original reference 0DCB: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5630–5630. Prerequisite pointer invoked at proof

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line(s) 5776; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D67: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5552–5552. Prerequisite pointer invoked at proof line(s) 5801; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O}_\mathcal{C})$ be a ringed site. Assume $\mathcal{C}$ has fibre products. Let $\{U_i \to X\}_{i\in I}$ be a covering in $\mathcal{C}$. For $i \in I$ let E_i be an object of $D(\mathcal{O}_{U_i})$ and for $i, j \in I$ let*

$$\rho_{ij} : E_i|_{\mathcal{C}/U_{ij}} \longrightarrow E_j|_{\mathcal{C}/U_{ij}}$$

be an isomorphism in $D(\mathcal{O}_{U_{ij}})$ where $U_{ij} = U_i \times_X U_j$. Assume

1. *the ρ_{ij} satisfy the cocycle condition on $U_i \times_X U_j \times_X U_k$ for all $i, j, k \in I$,*
2. *$\mathcal{E}xt^p_{\mathcal{O}_{U_i}}(E_i, E_i) = 0$ for all $p < 0$ and $i \in I$, and*
3. *there exists a $t \in \mathbf{Z}$ such that $H^p(E_i) = 0$ for $p < t$ and all $i \in I$.*

Then there exists a unique pair (E, ρ_i) where E is an object of $D(\mathcal{O}_X)$ and $\rho_i : E|_{U_i} \to E_i$ are isomorphisms in $D(\mathcal{O}_{U_i})$ compatible with the ρ_{ij}.

First proof. In this proof we deduce the lemma from the very general Lemma 0DCB. We urge the reader to look at the second proof in stead.

We may replace $\mathcal{C}$ with $\mathcal{C}/X$. Thus we may and do assume X is the final object of $\mathcal{C}$ and that $\mathcal{C}$ has all finite limits.

Let $\mathcal{B}$ be the full subcategory of $\mathcal{C}$ consisting of $U \in \mathrm{Ob}(\mathcal{C})$ such that there exists an $i(U) \in I$ and a morphism $a_U : U \to U_{i(U)}$. We denote $E_U = a_U^* E_{i(U)}$ in $D(\mathcal{O}_U)$ the pullback (restriction) of E_i via a_U. Given a morphism $a : U \to U'$ of $\mathcal{B}$ we obtain a morphism $(a_{U'} \circ a, a_U) : U \to U_{i(U')} \times_X U_{i(U)} = U_{i(U')i(U)}$ and hence an isomorphism

$$\rho_a : a^* E_{U'} = a^* a_{U'}^* E_{i(U')} \xrightarrow{(a_{U'} \circ a, a_U)^* \rho_{i(U')i(U)}} a_U^* E_{i(U)} = E_U$$

in $D(\mathcal{O}_U)$. The data $\mathcal{B}, E_U, \rho_a$ are as in Situation 0D67; the isomorphisms ρ_a satisfy the cocycle condition exactly because of condition (1) in the statement of the lemma (details omitted).

We are going to apply Lemma 0DCB with $\mathcal{B}$, E_U, ρ_a as above and with $\mathcal{D} = \mathcal{C}$ and $f : \mathcal{C} \to \mathcal{D}$ the identity morphism. Assumptions (1) and (2)(a) of Lemma 0DCB we have seen above. Assumption (2)(b) of Lemma 0DCB is clear. Assumption (2)(c) of Lemma 0DCB holds because $\{U_i \to X\}$ is a covering[1]. Assumption (3) of Lemma 0DCB holds because we have assumed the vanishing of all negative Ext sheaves of E_i which certainly implies that for any object U lying over U_i the negative self-Exts of $E_i|_U$ are zero. Assumption (4) of Lemma 0DCB holds because we have assumed the cohomology sheaves of each E_i are zero to the left of t.

We obtain a unique solution (E, ρ_U). Setting $\rho_i = \rho_{U_i}$ the lemma follows. □

[1] In fact, it would suffice if the map $\coprod_{i\in I} h_{U_i} \to h_X$ becomes surjective on sheafification and the lemma holds in this case with the same proof.